Improvement in Food Resources Class: 9^{th}

Textual Questions

ON PAGE.NO: 245

Q1: What do we get from cereals, pulses, fruits and vegetables?

Ans: The cereals provide us carbohydrate. The pulses give us protein. Fruits and vegetables give us a range of vitamins and minerals. In addition the fruits and vegetables give carbohydrates, proteins, fat and lots of fibres.

ON PAGE.NO: 247

Q1: How do biotic and abiotic factors affect crop production?

Ans: Under different situations crop suffers due to biotic stresses (such as diseases caused by bacteria, fungi, insects etc) and abiotic stresses (such as drought, salinity, heat, cold etc), which cause a great loss of production. If crop varieties are developed which are resistant to these stresses, then crop production can be improved significantly.

Q2: What are the desirable agronomic characteristics for crop improvement?

Ans: The desirable agronomic characteristics for crop improvement are high yield, dwarfness, early maturation etc. These characteristics are introduced in the food plants to increase their yield.

Q3: What are macronutrients and why are they called macronutrients?

Ans: The macronutrients are nitrogen, phosphorus, potassium, calcium, magnesium and sulphur. They are called macronutrients because they are required by crop plants in larger amounts.

Q4: How do plants get nutrients?

Ans: The plants get nutrients from air, water and soil. The inorganic elements e.g. nitrogen, phosphorus, calcium, magnesium, sulphur etc are obtained from soil in the form of minerals. The carbon and oxygen are obtained from air and the hydrogen from water.

ON PAGE.NO: 248

Q1: Compare the use of manure and fertilizers in maintaining soil fertility?

Ans: Manures contain many organic substances which can be easily degraded and absorbed by plants. It helps in recycling of biological wastes. They increase the fertility of soil for long duration without causing any harm. While as, fertilizers improve soil fertility for short duration but cause environmental hazards. Continuous use of fertilizers in a particular area causes destruction of soil fertility.

ON PAGE.NO: 251

Q1: Which of the following conditions will give the most benefits? Why?

a) Farmers use high quality seeds, do not adopt irrigation or use fertilizers.

b) Farmers use ordinary seeds, adopt irrigation and use fertilizers.

c) Farmers use quality seeds, adopt irrigation, use fertilizers and use crop protection measures.

Ans: The condition at (c) will give the most benefits because all these conditions are required for good crop production. High quality seeds germinate properly and grow to healthy plants. Irrigation helps crop plants to fight against draught stress. Fertilizers provide nutrients and crop protection measures plants from diseases.

Q2: Why should preventive measures and biological methods be preferred for protecting crops?

Ans: Preventive measures and biological methods should be preferred for protection of crops because they are ecologically safe, target specific and harmless to other life forms.

Q3: What factors may be responsible for losses of grains during storage?

Ans: There are two main factors responsible for losses during storage. They are 1) Abiotic, which includes temperature, light, water etc and 2) Biotic, which include insects, birds, bacteria etc.

ON PAGE.NO: 253

Q1: Which method is commonly used for improving cattle breeds and why?

Ans: Selective breeding is commonly used for improving cattle breeds. This is done by cross breeding the two different breeds which have useful characters, usually local breeds are cross breeded with exotic breeds. In this way, the desirable characters are transferred and thus obtained in next generation.

ON PAGE.NO: 254

Q1: Discuss the implications of the following statement;

"It is interesting to note that poultry is India's most efficient converter of low fibre food stuff (which is unfit for human consumption) into highly nutritious animal protein food".

Ans: Poultry birds utilize such agricultural by-products which are unfit for human consumption. On the other hand, they give us eggs and high quality meat which serves as a cheap source of animal protein.

Q2: What management practices are common in dairy and poultry farming?

Ans: These are:

I. Maintenance of temperature

II. Proper housing facilities having hygienic conditions
III. Proper feeding
IV. Preventive and control of diseases and pests

<u>Q3:</u> What are the differences between broilers and layers and in their management?

<u>Ans:</u>

<u>Layers</u>	<u>Broilers</u>
1. Layers are egg-laying birds, managed for the purpose of getting eggs.	1. Broilers are maintained for getting meat
2. Layers start producing eggs at the age of 20 weeks.	2. They are raised upto 6-7 weeks in poultry farms and then sent to market for meat purpose.
3. They require enough space and adequate lighting.	3. They require conditions to grow fast and low mortality.
4. They need restricted and calculated feed with vitamins, minerals and micronutrients .	4. The daily food requirement for broilers is rich in protein and vitamin K and A. The fat contents should also be adequate.

<u>ON PAGE.NO: 256</u>

<u>Q1:</u> How are fish obtained?

<u>Ans:</u> There are two methods of obtaining fish. One method is capture fishing in which fish are obtained from natural resources such as fresh water resources (I.e. canals, ponds, rivers etc), marine fishery resources (i.e. coastline, deep seas etc). The other method is by fish farming (or culture fishery) which is concerned with culturing, feeding, breeding and fish production.

Q2: What are the advantages of composite fish culture?

Ans: Composite fish culture is advantageous, economical and profitable from business point of view. It yields about 8-9 times more production as compared to monoculture.

ON PAGE.NO: 257

Q1: What are the desirable characters of bee varieties suitable for honey production?

Ans: The desirable characters of bee varieties suitable for honey production are:

1) They sting less.
2) They stay for longer period in a given bee hive.
3) They breed well.
4) They produce comparatively more honey and wax.

Q2: What is pasturage and how is it related to honey production?

Ans: Pasturage is the availability of flowers for nectar and pollen collection. The quality and taste of the honey is determined by the kind and quantity of pasturage.

ON PAGE NO: 259

Q1: Explain any one method of crop production which ensures high yield?

Ans: One method used for crop production which ensures high yield is plant breeding. It involves the improving of varieties of crops by breeding plants. The plants from different areas/places is picked up with desired traits and then hybridistsion or cross breeding of these varieties is done to obtain a plant/crop of desired characteristics.

<u>Q2:</u> Why are manure and fertilizers used in fields?

<u>Ans:</u> They are used to ensure vegetative growth (leaves, branches and flowers), giving rise to healthy plants, that result in high crop production.

<u>Q3:</u> What are the advantages of inter-cropping and crop rotation?

<u>Ans:</u> <u>Advantages of using inter-cropping:</u>

1) It helps to maintain soil fertility
2) It increases productivity per unit area
3) Save labour and time
4) Both crops can be easily harvested and processed separately

<u>Advantages of using crop rotation:</u>

1) It improves the soil fertility
2) It avoids depletion of a particular nutrient from soil
3) It minimize pest infestation and diseases
4) It helps in weed control
5) It prevents change in the chemical nature of the soil

<u>Q4:</u> What is genetic manipulation? How is it useful in agricultural practices?

<u>Ans:</u> Genetic manipulation is a process of incorporating desirable (genes) characters into crop varieties by hybridization. Hybridization involves crossing between genetically dissimilar plants. It is useful in developing varieties which shows:

- Increased yield
- Better quality
- Shorter and early maturity period
- Better adaptability to adverse environmental conditions
- Desirable characteristics

<u>Q5:</u> How do storage grain losses occur?

<u>Ans:</u> The factors responsible for loss of grains during storage are:

1) Abiotic factors like moisture, humidity, temperature etc
2) Biotic factors like insects, rodents, birds, bacteria etc

<u>Q6:</u> How do good animal husbandry practices benefit farmers?

<u>Ans:</u> Good animal husbandry practices are beneficial to the farmers in the following ways:

1) Improvement of breeds of the domesticated animals.
2) Increasing the yield of foodstuffs such as milk, eggs and meat.
3) Proper management of domestic animals in terms of shelter, feeding, care and protection against diseases
   These ultimately help the farmers to improve their economic condition.

<u>Q7:</u> What are the benefits of cattle farming?

<u>Ans:</u> Cattle farming is beneficial in the following ways:

1) Milk production is increased by high yielding animals.
2) Good quality of meat, fibre and skin can be obtained.
3) Good breed of draught animals can be obtained.

<u>Q8:</u> For increasing production, what is common in poultry, fisheries and bee-keeping?

<u>Ans:</u> Variety improvement, housing, rearing, sanitation, disease control and marketing.

<u>Q9:</u> How do you differentiate between capture fishing, mariculture, and aquaculture?

<u>Ans:</u> <u>Capture fishing:</u> It is the fishing in which fishes are captured from natural resources like pond, sea water etc

<u>Mariculture:</u> It is the culture of fish in marine water.

<u>Aquaculture:</u> It is done both in fresh water and in marine water.